

Jack DeGroot

Wantage, NJ | 862-354-1372 | degrootjack12@gmail.com | github.com/JackDeGroot

EDUCATION

University of New Hampshire, Durham, NH

- Masters in Statistics GPA: 3.9

Marywood University, Scranton, PA

- Bachelor of Science Mathematics and Finance GPA: 3.5

WORK EXPERIENCE

Adjunct Instructor, New Jersey Institute of Technology, Newark, NJ September 2024 – Current

- Coordinating and instructing two sections of MATH 105: Elementary Probability and Statistics, covering probability, distributions, and hypothesis testing. Managed course coordination across both sections, including curriculum alignment, grading standards, and student progress tracking.
- Taught one section of MATH 225: Survey of Probability and Statistics, covering foundational probability, descriptive statistics, and introductory statistical inference to undergraduate STEM students.
- Instructed six sections of MATH 111: Calculus I at NJIT, covering core concepts including limits, differentiation, and integration, and guiding students through practical applications in engineering.

Adjunct Lecturer, Rutgers University, Newark, NJ September 2025 – Current

- Instructed MATH 114 and MATH 135: both four-credit Precalculus and Calculus 1 course covering all varieties of functions, trigonometry, analytic geometry, as well as limits, derivatives, and integrals.
- Designed and administered all course assessments—including quizzes, exams, and a comprehensive final—for ~30 undergraduate students, ensuring mastery of concepts essential for success in Calculus.

Adjunct Instructor, University of New Hampshire, Durham, NH May 2022 – July 2022

- Coordinated and taught MATH 644: Statistics for Engineers and Scientists covering Design of Experiments, Exploratory Data Analysis, Probability and Probability Distributions, Statistical Inference including Hypothesis Testing, regression and correlation, and ANOVA.

PROFESSIONAL EXPERIENCE

Sussex County Commissioner - Elected Official January 2023 - Present

- Oversee a \$126M operating budget and \$13.7M capital budget, applying quantitative analysis and data-informed decision-making to evaluate program outcomes, performance metrics, and resource allocation.
- Initiated and developed two new partnerships between the County College and external organizations, aligning academic programs with community and industry workforce needs.
- Lead strategic planning, policy development, and risk evaluation across county departments, synthesizing complex operational data to guide high-stakes decisions affecting residents, businesses, and institutions.
- Serve as senior liaison to the County College, advising on strategic curriculum development and institutional direction

Office of New Hampshire Governor Christopher Sununu, Concord NH January 2023 - May 2023

- Conducted large-scale statistical analysis of survey data from over 500 schools statewide to evaluate the adoption and effectiveness of social-emotional learning programs.
- Identified patterns in curriculum integration and behavioral health program usage across diverse institutional contexts, translating complex datasets into actionable insights that informed executive-level strategic decisions.
- Delivered findings that shaped statewide policy on multi-tiered student support systems and school safety practices, demonstrating the ability to communicate statistical results clearly to non-technical stakeholders.

RESEARCH EXPERIENCE

Methods in Functional Data Analysis: Modeling Response Curves as Functions of Experimental Factors in Designed Experiments

January 2023 - September 2023

- Utilized Functional Principal Components, Bayesian Hierarchical two-stage mixed-effects model, and JMP's Fit Curve method to model functional response curves as direct functions of experimental factors.
- Recognized dynamic relationships between experimental factors and curve behavior to enhance predictive accuracy and optimization of experimental parameters.

A Mathematical Model for Forecasting the Spread of Covid-19 in PA

June 2020 - July 2021

- Worked with Cody Dosch, Heather Kwolek, and faculty advisor Dr. Craig Johnson to create a Susceptible, Vaccinated, Exposed, Infected, and Removed (SVEIR) mathematical model for the spread of Covid-19 that involved nonlinear systems of equations to help foresee the virus's spread in a set population.
- Identified preventative measures Marywood University should implement, methods of virus transmission, and proper precautions when a student tests positive for Covid-19 while living on Marywood's campus.

PRESENTATIONS

- Rose-Hulman Undergraduate Mathematics Conference, A Mathematical Model for Forecasting the Spread of Covid-19 in Pennsylvania, (Oral Presentation), April 23, 2021
- Annual Meeting of the Pennsylvania Academy of Science, A Mathematical Model for Forecasting the Spread of Covid-19 in Pennsylvania, (Oral Presentation), April 10, 2021
- Moravian College Student Mathematics Conference, A Mathematical Model for Forecasting the Spread of Covid-19 in Pennsylvania, (Oral Presentation), February 13, 2021

TECHNICAL SKILLS

- *Computer Languages:* R, MATLAB, Python, SAS, JMP, Excel
- *Tools:* ggplot2, dplyr, tidyr, knitr, Pandas, Numpy, Keras, Tensorflow